

STEAM-AI Diagnostic Report

STEAM_V322_2025, horizon 2025, run STEAM_V322_2025_2025_caf466c3

Generated 2026-09-22T22:30:01+00:00 by STEAM-AI 0.1.0. Ingested 2026-09-22T22:13:07.555746+00:00.

DRAFT: DRAFT - generated by STEAM-AI for modeller review

Run details

Item	Value
Run id	STEAM_V322_2025_2025_caf466c3
Scenario	STEAM_V322_2025
Horizon year	2025
Policy set	REF
STEAM version	3.2.2
Ingested at	2026-09-22T22:13:07.555746+00:00
Report generated at	2026-09-22T22:30:01+00:00
STEAM-AI version	0.1.0

[src: manifest.json]

This report lists what STEAM-AI found when it checked the run, ranked by how much each issue matters. Every number carries a source reference in small grey text. The report is a draft for modeller review and is not a model result.

Executive summary

Run health score: 57.0 out of 100, grade D.

[src: health.json]

Definition: Health = 0.7 x Findings component + 0.3 x Convergence component. Findings component = max(0, 100 - 25 x Critical - 8 x High - 2 x Medium - 0 x Info), counting each distinct problem type (a check and the issue it found) once, at its worst significant severity: a coding error repeated on 8,000 links is one problem. Convergence component = 100 if every assignment period met the relative gap target, scaled down linearly to 0 at ten times the target. When a run has no convergence data, the findings component carries the full weight. Grade: A >= 90, B >= 75, C >= 60, D >= 40, E below 40.

[src: config/health.yaml; health.json]

Health score components

Component	Score	Weight	Detail
Findings	57.0	1.00	100 - 43 penalty over 4 distinct problem type(s) across 30 significant finding(s) (Critical 6, High 22, Medium 2); carries full weight because convergence data is unavailable
Convergence	0.0	0.00	no convergence table in this run

[src: health.json]

The checks produced 30 findings: 6 Critical, 22 High, 2 Medium, 0 Info.

[src: findings.json]

Findings by severity

Severity	Findings	Significant
Critical	6	6
High	22	22
Medium	2	2
Info	0	0

[src: findings.json]

Top five findings

Top five findings by severity

Rank	Severity	Location	Finding
1	Critical	Link 38234 (LOC, 1024970-1025069)	Link 38234 (LOC) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.
2	Critical	Link 38173 (LOC, 1024876-1024970)	Link 38173 (LOC) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.
3	Critical	Link 38233 (LOC, 1024970-1024876)	Link 38233 (LOC) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.
4	Critical	Link 39138 (LOC, 1026735-1026715)	Link 39138 (LOC) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.
5	Critical	Link 38324 (LOC, 1025069-1024970)	Link 38324 (LOC) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.

[src: findings/299916ca236a; findings/321519d6ebb1; findings/5a439d89472e; findings/804b4dc2ece0; findings/a5ae16493c31]

What to fix first

28 Critical or High findings need action. Fix them in the order below; Critical items block use of the run.

[src: findings.json]

Action list

Severity	Location	Finding	Suggested action	Best measure
Critical	Link 38234 (LOC, 1024970-1025069)	Link 38234 (LOC) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.	Fix the link attributes or remove the link from the scenario network.	Repair the network coding (not computable, low confidence)
Critical	Link 38173 (LOC, 1024876-1024970)	Link 38173 (LOC) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.	Fix the link attributes or remove the link from the scenario network.	Repair the network coding (not computable, low confidence)
Critical	Link 38233 (LOC, 1024970-1024876)	Link 38233 (LOC) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.	Fix the link attributes or remove the link from the scenario network.	Repair the network coding (not computable, low confidence)
Critical	Link 39138 (LOC, 1026735-1026715)	Link 39138 (LOC) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.	Fix the link attributes or remove the link from the scenario network.	Repair the network coding (not computable, low confidence)
Critical	Link 38324 (LOC, 1025069-1024970)	Link 38324 (LOC) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.	Fix the link attributes or remove the link from the scenario network.	Repair the network coding (not computable, low confidence)
Critical	Link 39115 (LOC, 1026715-1026703)	Link 39115 (LOC) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.	Fix the link attributes or remove the link from the scenario network.	Repair the network coding (not computable, low confidence)
High	Island of 372 nodes (e.g. node 333155)	A group of 372 nodes is cut off from the rest of the network (main network: 59,582 nodes).	Add or fix the connecting link(s); trips to zones on the island cannot be assigned.	Repair the network coding (not computable, low confidence)
High	Island of 371 nodes (e.g. node 131138)	A group of 371 nodes is cut off from the rest of the network (main network: 59,582 nodes).	Add or fix the connecting link(s); trips to zones on the island cannot be assigned.	Repair the network coding (not computable, low confidence)
High	Island of 23 nodes (e.g. node 193907)	A group of 23 nodes meets the main network at the same point but under a different node number (node 1138164 sits on node 1138163).	Merge node 1138164 into node 1138163 (or renumber the links that use it) and re-check connectivity.	Repair the network coding (not computable, low confidence)
High	Island of 314 nodes (e.g. node 333115)	A group of 314 nodes is cut off from the rest of the network (main network: 59,582 nodes).	Add or fix the connecting link(s); trips to zones on the island cannot be assigned.	Repair the network coding (not computable, low confidence)

[src: findings/299916ca236a; findings/321519d6ebb1; findings/5a439d89472e; findings/804b4dc2ece0; findings/a5ae16493c31; findings/b32023c4fa16; findings/04703ce29df0; findings/053400e2c051; findings/12d8671a7d9d; findings/15066141cbb2]

18 further Critical or High findings are listed in the findings section.

[src: findings.json]

Findings by severity

Critical findings (6)

Network connectivity: Link 38234 (LOC, 1024970-1025069)

Critical: Link 38234 (LOC) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.

Modeller view: Method: attribute threshold on links

Values: issue=zero_capacity, capacity_vph=0, ffs_kph=30, lanes=0, link_class=LOC

Thresholds: min_capacity_vph=1, min_ffs_kph=5

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 38234

SQL: SELECT link_id, link_class, capacity_vph, ffs_kph, lanes, source_file, source_row FROM links WHERE capacity_vph < 1.0 OR ffs_kph < 5.0 ORDER BY link_id

[src: findings/299916ca236a; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38234 col capacity_vph]

Likely cause: Capacity or speed lookup missed this link class/area type, or the link was meant to be closed.

[src: findings/299916ca236a; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38234 col capacity_vph]

Suggested action: Fix the link attributes or remove the link from the scenario network.

[src: findings/299916ca236a; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38234 col capacity_vph]

Evidence

Metric	Value	Threshold
issue	zero_capacity	
capacity_vph	0	
ffs_kph	30	
lanes	0	
link_class	LOC	
min_capacity_vph		0
min_ffs_kph		5

[src: findings/299916ca236a; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38234 col capacity_vph]

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38234 col capacity_vph

[src: findings/299916ca236a; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38234 col capacity_vph]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/299916ca236a; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38234 col capacity_vph]

Network connectivity: Link 38173 (LOC, 1024876-1024970)

Critical: Link 38173 (LOC) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.

Modeller view: Method: attribute threshold on links

Values: issue=zero_capacity, capacity_vph=0, ffs_kph=30, lanes=0, link_class=LOC

Thresholds: min_capacity_vph=1, min_ffs_kph=5

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 38173

SQL: SELECT link_id, link_class, capacity_vph, ffs_kph, lanes, source_file, source_row FROM links WHERE capacity_vph < 1.0 OR ffs_kph < 5.0 ORDER BY link_id

[src: findings/321519d6ebb1; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38173 col capacity_vph]

Likely cause: Capacity or speed lookup missed this link class/area type, or the link was meant to be closed.

[src: findings/321519d6ebb1; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38173 col capacity_vph]

Suggested action: Fix the link attributes or remove the link from the scenario network.

[src: findings/321519d6ebb1; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38173 col capacity_vph]

Evidence

Metric	Value	Threshold
issue	zero_capacity	
capacity_vph	0	
ffs_kph	30	
lanes	0	
link_class	LOC	
min_capacity_vph		0
min_ffs_kph		5

[src: findings/321519d6ebb1; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38173 col capacity_vph]

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38173 col capacity_vph

[src: findings/321519d6ebb1; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38173 col capacity_vph]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/321519d6ebb1; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38173 col capacity_vph]

Network connectivity: Link 38233 (LOC, 1024970-1024876)

Critical: Link 38233 (LOC) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.

Modeller view: Method: attribute threshold on links

Values: issue=zero_capacity, capacity_vph=0, ffs_kph=30, lanes=0, link_class=LOC

Thresholds: min_capacity_vph=1, min_ffs_kph=5

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 38233

SQL: SELECT link_id, link_class, capacity_vph, ffs_kph, lanes, source_file, source_row FROM links WHERE capacity_vph < 1.0 OR ffs_kph < 5.0 ORDER BY link_id

[src: findings/5a439d89472e; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38233 col capacity_vph]

Likely cause: Capacity or speed lookup missed this link class/area type, or the link was meant to be closed.

[src: findings/5a439d89472e; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38233 col capacity_vph]

Suggested action: Fix the link attributes or remove the link from the scenario network.

[src: findings/5a439d89472e; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38233 col capacity_vph]

Evidence

Metric	Value	Threshold
issue	zero_capacity	
capacity_vph	0	
ffs_kph	30	
lanes	0	
link_class	LOC	
min_capacity_vph		0
min_ffs_kph		5

[src: findings/5a439d89472e; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38233 col capacity_vph]

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38233 col capacity_vph

[src: findings/5a439d89472e; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38233 col capacity_vph]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/5a439d89472e; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38233 col capacity_vph]

Network connectivity: Link 39138 (LOC, 1026735-1026715)

Critical: Link 39138 (LOC) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.

Modeller view: Method: attribute threshold on links

Values: issue=zero_capacity, capacity_vph=0, ffs_kph=30, lanes=0, link_class=LOC

Thresholds: min_capacity_vph=1, min_ffs_kph=5

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 39138

SQL: SELECT link_id, link_class, capacity_vph, ffs_kph, lanes, source_file, source_row FROM links WHERE capacity_vph < 1.0 OR ffs_kph < 5.0 ORDER BY link_id

[src: findings/804b4dc2ece0; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39138 col capacity_vph]

Likely cause: Capacity or speed lookup missed this link class/area type, or the link was meant to be closed.

[src: findings/804b4dc2ece0; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39138 col capacity_vph]

Suggested action: Fix the link attributes or remove the link from the scenario network.

[src: findings/804b4dc2ece0; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39138 col capacity_vph]

Evidence

Metric	Value	Threshold
issue	zero_capacity	
capacity_vph	0	
ffs_kph	30	
lanes	0	
link_class	LOC	
min_capacity_vph		0
min_ffs_kph		5

[src: findings/804b4dc2ece0; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39138 col capacity_vph]

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39138 col capacity_vph

[src: findings/804b4dc2ece0; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39138 col capacity_vph]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/804b4dc2ece0; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39138 col capacity_vph]

Network connectivity: Link 38324 (LOC, 1025069-1024970)

Critical: Link 38324 (LOC) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.

Modeller view: Method: attribute threshold on links

Values: issue=zero_capacity, capacity_vph=0, ffs_kph=30, lanes=0, link_class=LOC

Thresholds: min_capacity_vph=1, min_ffs_kph=5

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 38324

SQL: SELECT link_id, link_class, capacity_vph, ffs_kph, lanes, source_file, source_row FROM links WHERE capacity_vph < 1.0 OR ffs_kph < 5.0 ORDER BY link_id

[src: findings/a5ae16493c31; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38324 col capacity_vph]

Likely cause: Capacity or speed lookup missed this link class/area type, or the link was meant to be closed.

[src: findings/a5ae16493c31; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38324 col capacity_vph]

Suggested action: Fix the link attributes or remove the link from the scenario network.

[src: findings/a5ae16493c31; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38324 col capacity_vph]

Evidence

Metric	Value	Threshold
issue	zero_capacity	
capacity_vph	0	
ffs_kph	30	
lanes	0	
link_class	LOC	
min_capacity_vph		0
min_ffs_kph		5

[src: findings/a5ae16493c31; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38324 col capacity_vph]

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38324 col capacity_vph

[src: findings/a5ae16493c31; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38324 col capacity_vph]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/a5ae16493c31; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38324 col capacity_vph]

Network connectivity: Link 39115 (LOC, 1026715-1026703)

Critical: Link 39115 (LOC) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.

Modeller view: Method: attribute threshold on links

Values: issue=zero_capacity, capacity_vph=0, ffs_kph=30, lanes=0, link_class=LOC

Thresholds: min_capacity_vph=1, min_ffs_kph=5

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 39115

SQL: SELECT link_id, link_class, capacity_vph, ffs_kph, lanes, source_file, source_row FROM links WHERE capacity_vph < 1.0 OR ffs_kph < 5.0 ORDER BY link_id

[src: findings/b32023c4fa16; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39115 col capacity_vph]

Likely cause: Capacity or speed lookup missed this link class/area type, or the link was meant to be closed.

[src: findings/b32023c4fa16; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39115 col capacity_vph]

Suggested action: Fix the link attributes or remove the link from the scenario network.

[src: findings/b32023c4fa16; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39115 col capacity_vph]

Evidence

Metric	Value	Threshold
issue	zero_capacity	

Metric	Value	Threshold
capacity_vph	0	
ffs_kph	30	
lanes	0	
link_class	LOC	
min_capacity_vph		0
min_ffs_kph		5

[src: findings/b32023c4fa16; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39115 col capacity_vph]

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39115 col capacity_vph

[src: findings/b32023c4fa16; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39115 col capacity_vph]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/b32023c4fa16; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39115 col capacity_vph]

High findings (22)

Network connectivity: Island of 372 nodes (e.g. node 333155)

High: A group of 372 nodes is cut off from the rest of the network (main network: 59,582 nodes).

Modeller view: Method: union-find over undirected link graph

Values: issue=disconnected_component, component_size=372, gap_to_main_m=4.796e+04, fragment_node=1002762, nearest_main_node=501056, main_component_size=59582

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 8503, 8504, 8505, 8506, 8507, 8510, 8513, 8628, 30024, 30027 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/04703ce29df0; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8503 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8504 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8505 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8506 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8507 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8510 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8513 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8628 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30024 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30027 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30029 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30032 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30033 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30034 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30036 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30040 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30041 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30042 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30043 col link_id]

Likely cause: A missing link between the island and the main network, or a coding error in end-node ids.

[src: findings/04703ce29df0; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8503 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8504 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8505 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8506 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8507 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8510 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8513 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8628 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30024 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30027 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30029 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30032 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30033 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30034 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30036 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30040 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30041 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30042 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30043 col link_id]

Suggested action: Add or fix the connecting link(s); trips to zones on the island cannot be assigned.

Evidence

```
[src: findings/04703ce29df0; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8503 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8504 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8505 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8506 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8507 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8510 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8513 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8514 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8528 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30024 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30027 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30029 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30032 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30033 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30034 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30036 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30040 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30041 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30042 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30043 col link_id]
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[src: findings/04703ce29df0; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8503 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8504 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8505

col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8506 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8507 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8510 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8513 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8628 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30024 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30027 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30029 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30032 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30033 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30034 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30036 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30040 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30041 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30042 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30043 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/04703ce29df0; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8503 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8504 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8505 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8506 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8507 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8510 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8513 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8628 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30024 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30027 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30029 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30032 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30033 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30034 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30036 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30040 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30041 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30042 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 30043 col link_id]

Network connectivity: Island of 371 nodes (e.g. node 131138)

High: A group of 371 nodes is cut off from the rest of the network (main network: 59,582 nodes).

Modeller view: Method: union-find over undirected link graph

Values: issue=disconnected_component, component_size=371, gap_to_main_m=3881, fragment_node=1223142, nearest_main_node=3400016, main_component_size=59582

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 2851, 2855, 11440, 130038, 130039, 130101, 130102, 130114, 130115, 130159 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/053400e2c051; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2851 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2855 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11440 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130038 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130039 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130101 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130102 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130114 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130115 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130159 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130160 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130199 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130200 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130202 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130203 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130254 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130255 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130373 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130374 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130413 col link_id]

Likely cause: A missing link between the island and the main network, or a coding error in end-node ids.

[src: findings/053400e2c051; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2851 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2855 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11440 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130038 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130039 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130101 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130102 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130114 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130115 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130159 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130160 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130199 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130200 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130202 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130203 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130254 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130255 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130373 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130374 col link_id; network_2025_2040.bin.gz

(v322_AD_20250606_v19, 2025 fields) (links) row 130413 col link_id]

Suggested action: Add or fix the connecting link(s); trips to zones on the island cannot be assigned.

[src: findings/053400e2c051; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2851 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2855 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11440 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130038 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130039 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130101 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130102 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130114 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130115 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130159 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130160 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130199 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130200 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130202 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130203 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130254 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130255 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130373 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130374 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130413 col link_id]

Evidence

Metric	Value	Threshold
issue	disconnected_component	
component_size	371	
gap_to_main_m	3,881	
fragment_node	1,223,142	
nearest_main_node	3,400,016	
main_component_size	59,582	
example_nodes	20 items (full list in findings.json): 131,138, 131,148, 131,182	
example_links	20 items (full list in findings.json): 2,851, 2,855, 11,440	

[src: findings/053400e2c051; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2851 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2855 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11440 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130038 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130039 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130101 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130102 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130114 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130115 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130159 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130160 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130199 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130200 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130202 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130203 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130254 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130255 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130373 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130374 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130413 col link_id]

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2851 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2855 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11440 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130038 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130039 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130101 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130102 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130114 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130115 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130159 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130160 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130199 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130200 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130202 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130203 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130254 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130255 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130373 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130374 col link_id;

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130413 col link_id

[src: findings/053400e2c051; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2851 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2855 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11440 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130038 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130039 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130101 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130102 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130114 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130115 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130159 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130160 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130199 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130200 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130202 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130203 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130254 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130255 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130373 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130374 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130413 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/053400e2c051; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2851 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2855 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11440 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130038 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130039 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130101 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130102 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130114 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130115 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130159 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130160 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130199 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130200 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130202 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130203 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130254 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130255 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130373 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130374 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 130413 col link_id]

Network connectivity: Island of 23 nodes (e.g. node 193907)

High: A group of 23 nodes meets the main network at the same point but under a different node number (node 1138164 sits on node 1138163).

Modeller view: Method: union-find over undirected link graph

Values: issue=node_id_mismatch, component_size=23, gap_to_main_m=0, fragment_node=1138164, nearest_main_node=1138163, main_component_size=59582

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 4237, 88245, 88246, 88292, 88293, 88354, 88355, 88401, 88440, 88461 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/12d8671a7d9d; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4237 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88245 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88246 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88292 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88293 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88354 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88355 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88440 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88461 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88462 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88466 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88540 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88604 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88639 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88669 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88674 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88680 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88728 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88760 col link_id]

Likely cause: Two node numbers were coded at one location, so the links never join.

[src: findings/12d8671a7d9d; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4237 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88245 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88246 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88292 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88293 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88354 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88355 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88440 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88461 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88462 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88466 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88540 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88604 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88639 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88669 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88674 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88680 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88728 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88760 col link_id]

2025 fields) (links) row 88466 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88540 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88604 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88639 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88669 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88674 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88680 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88728 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88760 col link_id]

Suggested action: Merge node 1138164 into node 1138163 (or renumber the links that use it) and re-check connectivity.

[src: findings/12d8671a7d9d; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4237 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88245 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88246 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88292 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88293 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88354 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88355 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88440 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88461 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88462 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88540 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88604 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88639 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88669 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88674 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88680 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88728 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88760 col link_id]

Evidence

Metric	Value	Threshold
issue	node_id_mismatch	
component_size	23	
gap_to_main_m	0	
fragment_node	1,138,164	
nearest_main_node	1,138,163	
main_component_size	59,582	
example_nodes	20 items (full list in findings.json): 193,907, 193,908, 1,136,381	
example_links	20 items (full list in findings.json): 4,237, 88,245, 88,246	

[src: findings/12d8671a7d9d; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4237 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88245 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88246 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88292 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88293 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88354 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88355 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88440 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88461 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88462 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88540 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88604 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88639 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88669 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88674 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88680 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88728 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88760 col link_id]

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4237 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88245 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88246 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88292 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88293 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88354 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88355 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88440 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88461 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88462 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88466 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88540 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88604 col link_id;

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88639 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88669 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88674 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88680 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88728 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88760 col link_id

[src: findings/12d8671a7d9d; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4237 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88245 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88292 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88293 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88354 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88355 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88440 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88461 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88462 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88466 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88540 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88604 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88669 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88674 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88680 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88728 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88760 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/12d8671a7d9d; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4237 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88245 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88246 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88292 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88293 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88354 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88355 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88440 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88461 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88462 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88466 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88540 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88604 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88669 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88674 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88680 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88728 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 88760 col link_id]

Network connectivity: Island of 314 nodes (e.g. node 333115)

High: A group of 314 nodes is cut off from the rest of the network (main network: 59,582 nodes).

Modeller view: Method: union-find over undirected link graph

Values: issue=disconnected_component, component_size=314, gap_to_main_m=575.8,

fragment_node=1000655, nearest_main_node=334120, main_component_size=59582

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 8496, 8497, 8498, 8499, 8500, 8551, 8552, 8678, 8681, 28793 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/15066141cbb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8496 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8497 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8498 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8499 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8500 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8551 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8552 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8678 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8681 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28793 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28817 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28818 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28870 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28871 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28875 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28876 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28884 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28885 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28891 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28892 col link_id]

Likely cause: A missing link between the island and the main network, or a coding error in end-node ids.

[src: findings/15066141cbb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8496 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8497 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8498

Suggested action: Add or fix the connecting link(s); trips to zones on the island cannot be assigned.

Evidence

```
[src: findings/15066141cbb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8496 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8497 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8498 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8499 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8500 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8551 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8552 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8678 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8681 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8793 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28817 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28818 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28870 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28871 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28875 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28876 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28884 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28885 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28891 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28892 col link_id]
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STEAM-AI Diagnostic Report | run STEAM V322 2025-2025 caf466c3 | generated 2026-09-22T22:30:01+00:00 by STEAM-AI Page 15

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28870 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28871 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28875 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28876 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28884 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28885 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28891 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28892 col link_id

[src: findings/15066141cbb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8496 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8497 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8498 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8499 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8500 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8551 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8552 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8678 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8681 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28793 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28817 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28818 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28870 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28871 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28875 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28876 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28884 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28885 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28891 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28892 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/15066141cbb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8496 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8497 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8498 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8499 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8500 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8551 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8552 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8678 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8681 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28793 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28817 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28818 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28870 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28871 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28875 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28876 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28884 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28885 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28891 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28892 col link_id]

Network connectivity: 2,343 separate network fragments

High: 2,343 separate network fragments show the same problem: cut off from the main network. The 10 most severe are listed individually.

Modeller view: Method: 2,343 findings of issue 'disconnected_component' grouped; the 10 most severe kept as individual findings

Values: issue=disconnected_component, n_occurrences=2343, n_listed=10, by_severity={'High': 2343}, location_type=node, nodes_in_fragments=30773

Thresholds: group_issue_above=25, group_keep_worst=10

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 7, 17336, 8841, 11961, 11

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/24e130a98ce3; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17336 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11961 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11 col link_id]

Likely cause: A junction or connecting link is missing at the gap, so the fragment stops short of the main network.

[src: findings/24e130a98ce3; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17336 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11961 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11 col link_id]

Suggested action: Fix the coding rule that produces this pattern rather than each occurrence; add the connecting link between node 501316 and node 501299, or confirm the gap is intended.

[src: findings/24e130a98ce3; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17336 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11961 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11 col link_id]

Evidence

Metric	Value	Threshold
issue	disconnected_component	
n_occurrences	2,343	
n_listed	10	
by_severity	High=2,343	
location_type	node	
nodes_in_fragments	30,773	
example_ids	50 items (full list in findings.json): 52204, 412943, 334659	
summarised_ids	500 items (full list in findings.json): 200743, 196400, 1204290	
group_issue_above		25
group_keep_worst		10

[src: findings/24e130a98ce3; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17336 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11961 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11 col link_id]

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17336 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8841 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11961 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11 col link_id

[src: findings/24e130a98ce3; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17336 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11961 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/24e130a98ce3; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17336 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11961 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11 col link_id]

Network connectivity: Island of 495 nodes (e.g. node 52262)

High: A group of 495 nodes is cut off from the rest of the network (main network: 59,582 nodes).

Modeller view: Method: union-find over undirected link graph

Values: issue=disconnected_component, component_size=495, gap_to_main_m=78, fragment_node=501064, nearest_main_node=501024, main_component_size=59582

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 11, 12, 8535, 8544, 8545, 8686, 8687, 8693, 8694, 19487 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/27bb8b8034de; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 12 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8535 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8544 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8545 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8686 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8687 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8693 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8694 col link_id]

Likely cause: A missing link between the island and the main network, or a coding error in end-node ids.

Suggested action: Add or fix the connecting link(s); trips to zones on the island cannot be assigned.

Evidence

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[src: findings/27bb8b8034de; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 12 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8535 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8544 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8545 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8686 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8687 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8693 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8694 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19487 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19488 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19490 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19491 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19492 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19493 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19494 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19495 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19496 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19497 col link_id]
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STEAM-AI Diagnostic Report | run STEAM V322 2025 2025 caf466c3 | generated 2026-09-22T22:30:01+00:00 by STEAM-AI-Pipe-018

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8545 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8686 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8687 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8693 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8694 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19487 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19488 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19489 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19490 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19491 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19492 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19493 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19494 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19495 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19496 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19497 col link_id

[src: findings/27bb8b8034de; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 12 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8535 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8544 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8545 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8686 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8687 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8693 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8694 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19487 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19488 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19489 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19490 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19491 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19492 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19493 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19494 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19495 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19496 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19497 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/27bb8b8034de; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 12 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8535 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8544 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8545 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8686 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8687 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8693 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8694 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19487 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19488 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19489 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19490 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19491 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19492 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19493 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19494 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19495 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19496 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 19497 col link_id]

Network connectivity: Island of 618 nodes (e.g. node 386725)

High: A group of 618 nodes is cut off from the rest of the network (main network: 59,582 nodes); its nearest node (760706) is 24 m from main-network node 760707.

Modeller view: Method: union-find over undirected link graph

Values: issue=disconnected_component, component_size=618, gap_to_main_m=24.3, fragment_node=760706, nearest_main_node=760707, main_component_size=59582

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 11961, 11962, 11963, 11964, 11965, 11966, 11967, 11978, 11979, 11980 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/29d845063aa3; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11961 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11962 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11963 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11964 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11965 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11966 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11967 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11978 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11980 col link_id]

Likely cause: A junction or connecting link is missing at the gap, so the fragment stops short of the main network.

Suggested action: Add the connecting link between node 760706 and node 760707, or confirm the gap is intended.

Evidence

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[src: findings/29d845063aa3; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11961 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11962 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 11963 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11964 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11965 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 11966 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11967 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11978 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 11979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11980 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 11982 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11983 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11984 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 11985 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11986 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11987 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 11988 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11989 col link_id;
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network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11994 col link_id]

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11961 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11962 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11963 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11964 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11965 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11966 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11967 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11978 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11979 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11980 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11981 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11982 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11983 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11984 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11985 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11986 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11987 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11988 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11989 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11994 col link_id

[src: findings/29d845063aa3; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11961 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11962 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11963 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11964 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11965 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11966 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11967 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11978 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11980 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11982 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11983 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11984 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11985 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11986 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11987 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11988 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11989 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11994 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/29d845063aa3; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11961 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11962 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11963 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11964 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11965 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11966 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11967 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11978 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11980 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11982 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11983 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11984 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11985 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11986 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11987 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11988 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11989 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 11994 col link_id]

Network connectivity: Island of 270 nodes (e.g. node 104008)

High: A group of 270 nodes is cut off from the rest of the network (main network: 59,582 nodes); its nearest node (195809) is 45 m from main-network node 195805.

Modeller view: Method: union-find over undirected link graph

Values: issue=disconnected_component, component_size=270, gap_to_main_m=45,

fragment_node=195809, nearest_main_node=195805, main_component_size=59582

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 265, 268, 291, 2670, 2671, 2672, 2673, 4504, 4507, 4525 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/31ed4a16e659; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 265 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 268 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 291 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2670 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2671 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2672 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2673 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4504 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4507 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4525 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4536 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83268 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83272 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83295 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83296 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83316 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83317 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83449 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83450 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83451 col link_id]

Likely cause: A junction or connecting link is missing at the gap, so the fragment stops short of the main network.

[src: findings/31ed4a16e659; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 265 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 268 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 291 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2670 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2671 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2672 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2673 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4504 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4507 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4525 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4536 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83268 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83272 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83295 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83296 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83316 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83317 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83449 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83450 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83451 col link_id]

Suggested action: Add the connecting link between node 195809 and node 195805, or confirm the gap is intended.

[src: findings/31ed4a16e659; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 265 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 268 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 291 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2670 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2671 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2672 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2673 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4504 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4507 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4525 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4536 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83268 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83272 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83295 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83296 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83316 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83317 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83449 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83450 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83451 col link_id]

Evidence

Metric	Value	Threshold
issue	disconnected_component	
component_size	270	
gap_to_main_m	45	
fragment_node	195,809	
nearest_main_node	195,805	
main_component_size	59,582	
example_nodes	20 items (full list in findings.json): 104,008, 104,029, 104,052	
example_links	20 items (full list in findings.json): 265, 268, 291	

[src: findings/31ed4a16e659; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 265 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 268 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 291 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2670 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2671 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2672 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2673 col link_id; network_2025_2040.bin.gz

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 265 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 268 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 291 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2670 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2671 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2672 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2673 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4504 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4507 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4525 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4536 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83268 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83272 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83295 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83296 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83316 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83317 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83449 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83450 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83451 col link_id

Proposed measures

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[src: findings/31ed4a16e659; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 265 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 268 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 291 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2670 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2671 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2672 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 2673 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4504 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4507 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4525 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 4536 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83268 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83272 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83295 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83296 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83316 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83317 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83449 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83450 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 83451 col link_id]
```

ade023

High: A group of 25 nodes meets the main network at the same point but under a different node number (node 201343 sits on node 201344).

Modeller view: Method: union-find over undirected link graph

Values: issue=node_id_mismatch, component_size=25, gap_to_main_m=0, fragment_node=201343, nearest_main_node=201344, main_component_size=59582

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 7187, 128688, 128689, 128690, 128691, 128692, 128693, 128698, 128699, 129119 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/46ae046d6b57; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7187 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128688 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128689 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128690 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128691 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128692 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128693 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128698 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128699 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129119 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129120 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129121 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129122 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129123 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129127 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129128 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129129 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129130 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129131 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129132 col link_id]

Likely cause: Two node numbers were coded at one location, so the links never join.

[src: findings/46ae046d6b57; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7187 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128688 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128689 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128690 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128691 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128692 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128693 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128698 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128699 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129119 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129120 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129121 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129122 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129123 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129127 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129128 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129129 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129130 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129131 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129132 col link_id]

Suggested action: Merge node 201343 into node 201344 (or renumber the links that use it) and re-check connectivity.

[src: findings/46ae046d6b57; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7187 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128688 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128689 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128690 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128691 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128692 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128693 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128698 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 128699 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129119 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129120 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129121 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129122 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129123 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129127 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129128 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129129 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129130 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129131 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129132 col link_id]

Evidence

Metric	Value	Threshold
issue	node_id_mismatch	
component_size	25	
gap_to_main_m	0	
fragment_node	201,343	
nearest_main_node	201,344	
main_component_size	59,582	
example_nodes	20 items (full list in findings.json): 201,342, 201,343, 201,349	

(v322_AD_20250606_v19, 2025 fields) (links) row 129121 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129122 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129123 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129127 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129128 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129129 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129130 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129131 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 129132 col link_id]

Network connectivity: Island of 23 nodes (e.g. node 1075499)

High: A group of 23 nodes meets the main network at the same point but under a different node number (node 6272157 sits on node 1075490).

Modeller view: Method: union-find over undirected link graph
Values: issue=node_id_mismatch, component_size=23, gap_to_main_m=0, fragment_node=6272157, nearest_main_node=1075490, main_component_size=59582
Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 63078, 63079, 63082, 63083, 63092, 63093, 63124, 63125, 63288, 63295 (+10 more)
SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL
[src: findings/4ab156e86506; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63078 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63079 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63082 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63083 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63092 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63093 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63124 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63125 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63288 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63295 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63296 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63332 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63354 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63355 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63356 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63518 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63554 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63566 col link_id]

Likely cause: Two node numbers were coded at one location, so the links never join.

[src: findings/4ab156e86506; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63078 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63079 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63082 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63083 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63092 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63093 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63124 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63125 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63288 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63295 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63296 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63331 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63332 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63354 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63355 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63356 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63518 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63554 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63566 col link_id]

Suggested action: Merge node 6272157 into node 1075490 (or renumber the links that use it) and re-check connectivity.

[src: findings/4ab156e86506; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63078 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63079 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63082 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63083 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63092 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63093 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63124 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63125 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63288 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63295 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63296 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63331 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63332 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63354 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63355 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63356 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63518 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63554 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63566 col link_id]

Evidence

Metric	Value	Threshold
issue	node_id_mismatch	
component_size	23	

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/4ab156e86506; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63078 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63079 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63082 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63083 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63092 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63093 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63124 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63125 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63288 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63295 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63296 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63331 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63332 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63354 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63355 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63356 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63518 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63554 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63565 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 63566 col link_id]

Network connectivity: Island of 314 nodes (e.g. node 200552)

High: A group of 314 nodes is cut off from the rest of the network (main network: 59,582 nodes); its nearest node (202375) is 38 m from main-network node 202377.

Modeller view: Method: union-find over undirected link graph

Values: issue=disconnected_component, component_size=314, gap_to_main_m=37.6,

fragment_node=202375, nearest_main_node=202377, main_component_size=59582

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 6833, 6834, 6835, 6836, 6837, 6838, 6839, 6840, 6841, 6843 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/4e986b32209b; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6833 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6834 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6835 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6836 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6837 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6838 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6839 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6840 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6843 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6846 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6950 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6951 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6954 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6955 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6956 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6957 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6958 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7208 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7209 col link_id]

Likely cause: A junction or connecting link is missing at the gap, so the fragment stops short of the main network.

[src: findings/4e986b32209b; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6833 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6834 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6835 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6836 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6837 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6838 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6839 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6840 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6843 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6846 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6950 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6951 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6954 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6955 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6956 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6957 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6958 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7208 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7209 col link_id]

Suggested action: Add the connecting link between node 202375 and node 202377, or confirm the gap is intended.

[src: findings/4e986b32209b; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6833 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6834 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6835 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6836 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6837 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6838 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6839 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6840 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6843 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6846 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6950 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6951 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6954 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6955 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6956 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6957 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6958 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7208 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7209 col link_id]

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6954 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6955 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6956 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6957 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6958 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7208 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7209 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/4e986b32209b; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6833 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6834 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6835 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6836 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6837 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6838 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6839 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6840 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6843 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6846 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6950 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6951 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6954 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6955 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6956 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6957 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6958 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7208 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7209 col link_id]

Network connectivity: Island of 63 nodes (e.g. node 202703)

High: A group of 63 nodes meets the main network at the same point but under a different node number (node 1243673 sits on node 1243671).

Modeller view: Method: union-find over undirected link graph

Values: issue=node_id_mismatch, component_size=63, gap_to_main_m=0, fragment_node=1243673, nearest_main_node=1243671, main_component_size=59582

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 7556, 7559, 140834, 140835, 140845, 140846, 140891, 140892, 140922, 140923 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/70cc58836fb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7559 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140834 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140835 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140845 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140846 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140891 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140892 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140923 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140964 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140980 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140999 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141000 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141001 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141002 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141003 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141004 col link_id]

Likely cause: Two node numbers were coded at one location, so the links never join.

[src: findings/70cc58836fb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7559 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140834 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140835 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140845 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140846 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140891 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140892 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140923 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140964 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140980 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140999 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141000 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141001 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141002 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141003 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141004 col link_id]

Suggested action: Merge node 1243673 into node 1243671 (or renumber the links that use it) and re-check connectivity.

Evidence

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src: findings/70cc58836fb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7559 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140834 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140835 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140845 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140846 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140891 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140892 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140923 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140964 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140980 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140999 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141000 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141001 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141002 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141003 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141004 col link_id]
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STEAM-AI Diagnostic Report | run STEAM V322 2025 2025 caf466c3 | generated 2026-09-22T22:30:01+00:00 by STEAM-AI Page 031

[src: findings/70cc58836fb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7559 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140834 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140835 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140845 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140846 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140891 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140892 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140923 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140964 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140980 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140999 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141000 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141001 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141002 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141003 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141004 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/70cc58836fb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7559 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140834 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140835 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140845 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140846 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140891 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140892 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140923 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140964 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140980 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 140999 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141000 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141001 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141002 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141003 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 141004 col link_id]

Network connectivity: Island of 32 nodes (e.g. node 202361)

High: A group of 32 nodes meets the main network at the same point but under a different node number (node 202376 sits on node 202377).

Modeller view: Method: union-find over undirected link graph
Values: issue=node_id_mismatch, component_size=32, gap_to_main_m=0, fragment_node=202376, nearest_main_node=202377, main_component_size=59582
Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 7336, 135741, 135874, 135951, 135954, 135973, 135974, 135981, 135982, 135983 (+10 more)
SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/8a76158e1013; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7336 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135741 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135874 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135951 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135954 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135973 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135974 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135982 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135983 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135992 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136018 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136019 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136029 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136030 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136063 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136064 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136069 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136085 col link_id]

Likely cause: Two node numbers were coded at one location, so the links never join.

[src: findings/8a76158e1013; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7336 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135741 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135874 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135951 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135954 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135973 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135974 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135982 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135983 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135992 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136018 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136019 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136029 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136030 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136063 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136064 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136069 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136085 col link_id]

(v322_AD_20250606_v19, 2025 fields) (links) row 136029 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136030 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136063 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136064 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136069 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136085 col link_id]

Suggested action: Merge node 202376 into node 202377 (or renumber the links that use it) and re-check connectivity.

[src: findings/8a76158e1013; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7336 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135741 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135874 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135951 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135954 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135973 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135974 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135982 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135983 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135992 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136018 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136019 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136029 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136030 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136063 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136064 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136069 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136085 col link_id]

Evidence

Metric	Value	Threshold
issue	node_id_mismatch	
component_size	32	
gap_to_main_m	0	
fragment_node	202,376	
nearest_main_node	202,377	
main_component_size	59,582	
example_nodes	20 items (full list in findings.json): 202,361, 202,376, 1,232,983	
example_links	20 items (full list in findings.json): 7,336, 135,741, 135,874	

[src: findings/8a76158e1013; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7336 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135741 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135874 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135951 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135954 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135973 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135974 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135982 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135983 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135992 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136018 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136019 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136029 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136030 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136063 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136064 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136069 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136085 col link_id]

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7336 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135741 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135874 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135951 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135954 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135973 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135974 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135982 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135983 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135992 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136018 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136019 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136029 col link_id;

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136030 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136063 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136064 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136069 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136085 col link_id

[src: findings/8a76158e1013; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7336 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135741 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 135874 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
135951 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135954 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 135973 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
135974 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135981 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 135982 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
135983 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135992 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 136018 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
136019 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136028 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 136029 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
136030 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136063 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 136064 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
136069 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136085 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/8a76158e1013; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7336 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135741 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 135874 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
135951 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135954 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 135973 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
135974 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135981 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 135982 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
135983 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 135992 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 136018 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
136019 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136028 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 136029 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
136030 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136063 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 136064 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
136069 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 136085 col link_id]

Network connectivity: Island of 1410 nodes (e.g. node 412943)

High: A group of 1,410 nodes is cut off from the rest of the network (main network: 59,582 nodes); its nearest node (1191628) is 10 m from main-network node 1191627.

Modeller view: Method: union-find over undirected link graph

Values: issue=disconnected_component, component_size=1410, gap_to_main_m=10.1,

fragment_node=1191628, nearest_main_node=1191627, main_component_size=59582

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 17336, 17337, 17374, 17375, 27741, 27747, 27748, 27749, 27753, 27754 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/93a1f16ca66b; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17336 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17337 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 17374 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17375 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27741 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 27747 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27748 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27749 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 27753 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27754 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27755 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 28011 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108914 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108915 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 108921 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
108922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108935 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 108936 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
108952 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108953 col link_id]

Likely cause: A junction or connecting link is missing at the gap, so the fragment stops short of the main network.

Suggested action: Add the connecting link between node 1191628 and node 1191627, or confirm the gap is intended.

Evidence

```
[src: findings/93a1f16ca66b; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17336 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17337 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 17374 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17375 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27741 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 27747 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27748 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27749 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 27753 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27754 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27755 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 28011 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108914 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108915 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 108921 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
108922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108935 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 108936 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
108952 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108953 col link_id]
```

STEAM-AI Diagnostic Report | run STEAM V322 2025 2025 caf466c3 | generated 2026-09-22T22:30:01+00:00 by STEAM-AI Page 035

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27754 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27755 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28011 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108914 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108915 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108921 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108922 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108935 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108936 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108952 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108953 col link_id

[src: findings/93a1f16ca66b; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17336 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17337 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17374 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17375 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27741 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27747 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27748 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27749 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27753 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27754 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27755 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28011 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108914 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108915 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108921 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108935 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108936 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108952 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108953 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/93a1f16ca66b; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17336 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17337 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17374 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 17375 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27741 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27747 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27748 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27749 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27753 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27754 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27755 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 28011 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108914 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108915 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108921 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108935 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108936 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108952 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 108953 col link_id]

Network connectivity: Island of 25 nodes (e.g. node 901407)

High: A group of 25 nodes meets the main network at the same point but under a different node number (node 1196318 sits on node 1196320).

Modeller view: Method: union-find over undirected link graph

Values: issue=node_id_mismatch, component_size=25, gap_to_main_m=0, fragment_node=1196318, nearest_main_node=1196320, main_component_size=59582

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 27805, 113419, 113420, 113483, 113484, 113542, 113543, 113555, 113556, 113568 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/b6462f4acefa; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27805 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113419 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113420 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113483 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113484 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113542 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113543 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113555 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113568 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113569 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113575 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113576 col link_id; network_2025_2040.bin.gz

Likely cause: Two node numbers were coded at one location, so the links never join.

Suggested action: Merge node 1196318 into node 1196320 (or renumber the links that use it) and re-check connectivity.

Evidence

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[src: findings/b6462f4acefa; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27805 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113419 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113420 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113483 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113484 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113542 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113543 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113555 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113568 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113569 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113575 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113576 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113577 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113578 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113579 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113599 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113600 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113603 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113604 col link_id]
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STEAM-AI Diagnostic Report | run STEAM V322 2025 2025 caf466c3 | generated 2026-09-22T22:30:01+00:00 by STEAM-AI Page 037

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113483 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113484 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113542 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113543 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113555 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113556 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113568 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113569 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113575 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113576 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113577 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113578 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113579 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113599 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113600 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113603 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113604 col link_id

[src: findings/b6462f4acefa; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27805 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113419 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113420 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113483 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113484 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113542 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113543 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113555 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113568 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113569 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113575 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113576 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113577 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113578 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113579 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113599 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113600 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113603 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113604 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/b6462f4acefa; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27805 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113419 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113420 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113483 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113484 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113542 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113543 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113555 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113568 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113569 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113575 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113576 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113577 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113578 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113579 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113599 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113600 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113603 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 113604 col link_id]

Network connectivity: Island of 2722 nodes (e.g. node 52204)

High: A group of 2,722 nodes is cut off from the rest of the network (main network: 59,582 nodes); its nearest node (501316) is 11 m from main-network node 501299.

Modeller view: Method: union-find over undirected link graph

Values: issue=disconnected_component, component_size=2722, gap_to_main_m=11.1,

fragment_node=501316, nearest_main_node=501299, main_component_size=59582

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 7, 8, 9, 10, 6461, 6462, 6463, 8220, 8221, 8222 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

Likely cause: A junction or connecting link is missing at the gap, so the fragment stops short of the main network.

Suggested action: Add the connecting link between node 501316 and node 501299, or confirm the gap is intended.

Evidence

[src: findings/b782b172f2e5; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 9 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 10 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6461 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6462 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6463 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8220 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8222 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8223 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8224 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8225 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8227 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8228 col link_id; network_2025_2040.bin.gz

(v322_AD_20250606_v19, 2025 fields) (links) row 8229 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8230 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8232 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8233 col link_id]

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 9 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 10 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6461 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6462 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6463 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8220 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8221 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8222 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8223 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8224 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8225 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8226 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8227 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8228 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8229 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8230 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8232 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8233 col link_id

[src: findings/b782b172f2e5; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 9 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 10 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6461 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6462 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6463 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8220 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8222 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8223 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8224 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8225 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8227 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8228 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8229 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8230 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8232 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8233 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/b782b172f2e5; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 9 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 10 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6461 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6462 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 6463 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8220 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8222 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8223 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8224 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8225 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8227 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8228 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8229 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8230 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8232 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8233 col link_id]

Network connectivity: Island of 41 nodes (e.g. node 1217101)

High: A group of 41 nodes meets the main network at the same point but under a different node number (node 1217101 sits on node 1217085).

Modeller view: Method: union-find over undirected link graph

Values: issue=node_id_mismatch, component_size=41, gap_to_main_m=0, fragment_node=1217101, nearest_main_node=1217085, main_component_size=59582

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 126342, 126369, 126395, 126396, 126408, 126409, 126415, 126448, 126449, 126535 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/ce7203cf1732; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126342 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126369 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126395 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126396 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126408 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126409 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126415 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126448 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126449 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126535 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126536 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126540 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126541 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126551 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126552 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126571 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126572 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126584 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126585 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126587 col link_id]

Likely cause: Two node numbers were coded at one location, so the links never join.

[src: findings/ce7203cf1732; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126342 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126369 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126395 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126396 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126408 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126409 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126415 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126448 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126449 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126535 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126536 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126540 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126541 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126551 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126552 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126571 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126572 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126584 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126585 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126587 col link_id]

Suggested action: Merge node 1217101 into node 1217085 (or renumber the links that use it) and re-check connectivity.

[src: findings/ce7203cf1732; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126342 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126369 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126395 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126396 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126408 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126409 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126415 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126448 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126449 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126535 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126536 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126540 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126541 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126551 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126552 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126571 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126572 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126584 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126585 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126587 col link_id]

Evidence

Metric	Value	Threshold
issue	node_id_mismatch	
component_size	41	
gap_to_main_m	0	
fragment_node	1,217,101	
nearest_main_node	1,217,085	
main_component_size	59,582	
example_nodes	20 items (full list in findings.json): 1,217,101, 1,217,139, 1,217,148	
example_links	20 items (full list in findings.json): 126,342, 126,369, 126,395	

[src: findings/ce7203cf1732; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126342 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126369 col link_id; network_2025_2040.bin.gz

Network connectivity: Island of 17 nodes (e.g. node 1133139)

High: A group of 17 nodes meets the main network at the same point but under a different node number (node 1134244 sits on node 1134247).

Modeller view: Method: union-find over undirected link graph

Values: issue=node_id_mismatch, component_size=17, gap_to_main_m=0, fragment_node=1134244,

nearest_main_node=1134247, main_component_size=59582

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 86682, 86683, 86717, 86729, 86730, 86769, 86770, 86787, 86788, 86798 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/dc6a333c3c58; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86682 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86683 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86729 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86730 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86769 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86770 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86787 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86788 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86798 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86799 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86809 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86810 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86847 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86848 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86896 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86897 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86925 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86995 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 87004 col link_id]

Likely cause: Two node numbers were coded at one location, so the links never join.

[src: findings/dc6a333c3c58; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86682 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86683 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86729 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86730 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86769 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86770 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86787 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86788 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86798 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86799 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86809 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86810 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86847 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86848 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86896 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86897 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86925 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86995 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 87004 col link_id]

Suggested action: Merge node 1134244 into node 1134247 (or renumber the links that use it) and re-check connectivity.

[src: findings/dc6a333c3c58; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86682 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86683 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86729 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86730 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86769 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86770 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86787 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86788 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86798 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86799 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86809 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86810 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86847 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86848 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86896 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86897 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86925 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86995 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 87004 col link_id]

Evidence

Metric	Value	Threshold
issue	node_id_mismatch	
component_size	17	
gap_to_main_m	0	
fragment_node	1,134,244	
nearest_main_node	1,134,247	
main_component_size	59,582	

2025 fields) (links) row 86788 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86798 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86799 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86809 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86810 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86847 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86848 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86896 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86897 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86925 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 86995 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 87004 col link_id]

Network connectivity: Island of 631 nodes (e.g. node 334659)

High: A group of 631 nodes is cut off from the rest of the network (main network: 59,582 nodes); its nearest node (501052) is 15 m from main-network node 501044.

Modeller view: Method: union-find over undirected link graph

Values: issue=disconnected_component, component_size=631, gap_to_main_m=15, fragment_node=501052, nearest_main_node=501044, main_component_size=59582

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 8841, 8935, 13175, 13176, 13177, 13178, 13187, 13188, 13189, 13190 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/de7d721361f7; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8935 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13175 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13176 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13177 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13178 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13187 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13188 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13189 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13190 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13191 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13192 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13193 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13194 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13195 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13196 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13197 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13210 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13232 col link_id]

Likely cause: A junction or connecting link is missing at the gap, so the fragment stops short of the main network.

[src: findings/de7d721361f7; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8935 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13175 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13176 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13177 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13178 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13187 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13188 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13189 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13190 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13191 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13192 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13193 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13194 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13195 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13196 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13197 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13210 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13232 col link_id]

Suggested action: Add the connecting link between node 501052 and node 501044, or confirm the gap is intended.

[src: findings/de7d721361f7; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8935 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13175 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13176 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13177 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13178 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13187 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13188 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13189 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13190 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13191 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13192 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13193 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13194 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13195 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13196 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13197 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13210 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13232 col link_id]

Evidence

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/de7d721361f7; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 8935 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13175 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13176 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13177 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13178 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13187 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13188 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13189 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13190 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13191 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13192 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13193 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13194 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13195 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13196 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13197 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13210 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 13232 col link_id]

Network connectivity: Island of 111 nodes (e.g. node 725210)

High: A group of 111 nodes meets the main network at the same point but under a different node number (node 1151491 sits on node 1151474).

Modeller view: Method: union-find over undirected link graph

Values: issue=node_id_mismatch, component_size=111, gap_to_main_m=0, fragment_node=1151491, nearest_main_node=1151474, main_component_size=59582

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 95401, 95510, 95717, 95722, 95899, 95900, 95915, 95922, 96006, 96007 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/ec678e63546f; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95510 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95722 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95899 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95900 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95915 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96006 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96007 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96062 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96066 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96172 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96175 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96222 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96227 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96234 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96239 col link_id]

Likely cause: Two node numbers were coded at one location, so the links never join.

[src: findings/ec678e63546f; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95510 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95722 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95899 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95900 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95915 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96006 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96007 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96062 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96066 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96172 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96175 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96222 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96227 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96234 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96239 col link_id]

Suggested action: Merge node 1151491 into node 1151474 (or renumber the links that use it) and re-check connectivity.

[src: findings/ec678e63546f; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95510 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95722 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95899 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95900 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95915 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96006 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96007 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96062 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96066 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96172 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96175 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96222 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96227 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96234 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96239 col link_id]

Evidence

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[src: findings/ec678e63546f; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95401 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95510 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 95717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95722 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95899 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 95900 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95915 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 96006 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96007 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96062 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 96066 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96172 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96175 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 96221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96222 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2025 fields) (links) row 96227 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96234 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96239 col link_id]
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[src: findings/ec678e63546f; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95510 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95722 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95899 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95900 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95915 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96006 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96007 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96062 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96066 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96172 col link_id;

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96175 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96222 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96227 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96234 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96239 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/ec678e63546f; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95510 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95722 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95899 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95900 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95915 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96006 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96007 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96062 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96066 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96172 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96175 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96222 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96227 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96234 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 96239 col link_id]

Network connectivity: 73 network fragments

High: 73 network fragments show the same problem: touching the main network under a different node number. The 10 most severe are listed individually.

Modeller view: Method: 73 findings of issue 'node_id_mismatch' grouped; the 10 most severe kept as individual findings

Values: issue=node_id_mismatch, n_occurrences=73, n_listed=10, by_severity={'High': 73}, location_type=node, nodes_in_fragments=767

Thresholds: group_issue_above=25, group_keep_worst=10

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 95401, 7556, 126342, 7336, 27805

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/f9543520fd84; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126342 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7336 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27805 col link_id]

Likely cause: Two node numbers were coded at one location, so the links never join.

[src: findings/f9543520fd84; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126342 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7336 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27805 col link_id]

Suggested action: Fix the coding rule that produces this pattern rather than each occurrence; merge node 1151491 into node 1151474 (or renumber the links that use it) and re-check connectivity.

[src: findings/f9543520fd84; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126342 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7336 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27805 col link_id]

Evidence

Metric	Value	Threshold
issue	node_id_mismatch	
n_occurrences	73	
n_listed	10	
by_severity	High=73	
location_type	node	

Metric	Value	Threshold
nodes_in_fragments	767	
example_ids	50 items (full list in findings.json): 725210, 202703, 1217101	
summarised_ids	63 items (full list in findings.json): 200346, 1111641, 1200637	
group_issue_above		25
group_keep_worst		10

[src: findings/f9543520fd84; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126342 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7336 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27805 col link_id]

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126342 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7336 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27805 col link_id

[src: findings/f9543520fd84; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126342 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7336 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27805 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/f9543520fd84; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 126342 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7336 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 27805 col link_id]

Network connectivity: Island of 17 nodes (e.g. node 202429)

High: A group of 17 nodes meets the main network at the same point but under a different node number (node 1210377 sits on node 1210379).

Modeller view: Method: union-find over undirected link graph

Values: issue=node_id_mismatch, component_size=17, gap_to_main_m=0, fragment_node=1210377, nearest_main_node=1210379, main_component_size=59582

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 7381, 121607, 121608, 121656, 121657, 121695, 121696, 121706, 121707, 121709 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/fb1515052016; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7381 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121607 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121608 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121656 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121657 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121695 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121696 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121706 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121707 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121709 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121710 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121720 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121721 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121736 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121737 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121748 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121759 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121760 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121799 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121800 col link_id]

Likely cause: Two node numbers were coded at one location, so the links never join.

[src: findings/fb1515052016; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7381 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121607 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121608 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121656 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121657 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121720 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121721 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121736 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121737 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121748 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121759 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121760 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121799 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121800 col link_id

[src: findings/fb1515052016; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7381 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121607 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121608 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121656 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121657 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121695 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121696 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121706 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121707 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121709 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121710 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121720 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121721 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121736 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121737 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121748 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121759 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121760 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121799 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121800 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/fb1515052016; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 7381 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121607 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121608 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121656 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121657 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121695 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121696 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121706 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121707 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121709 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121710 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121720 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121721 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121736 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121737 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121748 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121759 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121760 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121799 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 121800 col link_id]

Medium findings (2)

Null values and identifier integrity: Link 38768 (UNK, 1025989-1026183)

Medium: 14 of 136,509 rows in the links table (0.0%) have no value for ffs_kph.

Modeller view: Method: COUNT(*) FILTER (WHERE column IS NULL) per required column

Values: table=links, column=ffs_kph, null_rows=14, rows=136509, null_share=0.0001026,

issue=null_values

Thresholds: null_share_high=0.01, null_share_medium=0

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 38768, 38848, 38849, 39135, 39192, 39193, 39195, 39296, 39313, 39314 (+4 more)

SQL: SELECT link_id, source_file, source_row FROM links WHERE "ffs_kph" IS NULL LIMIT 20

[src: findings/1d78acf16993; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38768 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38848 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38849 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39135 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39192 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39193 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39195 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39296 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39313 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39314 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39342 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39343 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row

153265 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153268 col ffs_kph]

Likely cause: Export mapping dropped the column for some records, or the network/matrix file has blank attributes.

[src: findings/1d78acf16993; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38768 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38848 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38849 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39135 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39192 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39193 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39195 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39296 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39313 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39314 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39342 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39343 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153265 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153268 col ffs_kph]

Suggested action: Fill or default ffs_kph in the source file and re-export; downstream checks skip rows with nulls.

[src: findings/1d78acf16993; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38768 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38848 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38849 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39135 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39192 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39193 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39195 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39296 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39313 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39314 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39342 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39343 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153265 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153268 col ffs_kph]

Evidence

Metric	Value	Threshold
table	links	
column	ffs_kph	
null_rows	14	
rows	136,509	
null_share	0.0%	
example_ids	14 items (full list in findings.json): 38,768, 38,848, 38,849	
issue	null_values	
null_share_high		1.0%
null_share_medium		0.0%

[src: findings/1d78acf16993; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38768 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38848 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38849 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39135 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39192 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39193 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39195 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39296 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39313 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39314 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39342 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39343 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153265 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153268 col ffs_kph]

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38768 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38848 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38849 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39135 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39192 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39193 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39195 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39296 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39313 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39314 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39342 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39343 col ffs_kph;

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153265 col ffs_kph;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153268 col ffs_kph

[src: findings/1d78acf16993; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38768 col ffs_kph;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38848 col ffs_kph; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 38849 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
39135 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39192 col ffs_kph; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 39193 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
39195 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39296 col ffs_kph; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 39313 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
39314 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39342 col ffs_kph; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 39343 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row
153265 col ffs_kph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153268 col ffs_kph]

Null values and identifier integrity: Link 38768 (UNK, 1025989-1026183)

Medium: 14 of 136,509 rows in the links table (0.0%) have no value for capacity_vph.

Modeller view: Method: COUNT(*) FILTER (WHERE column IS NULL) per required column
Values: table=links, column=capacity_vph, null_rows=14, rows=136509, null_share=0.0001026,
issue=null_values

Thresholds: null_share_high=0.01, null_share_medium=0

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) rows 38768, 38848, 38849,
39135, 39192, 39193, 39195, 39296, 39313, 39314 (+4 more)

SQL: SELECT link_id, source_file, source_row FROM links WHERE "capacity_vph" IS NULL LIMIT 20

[src: findings/3f089cf9e215; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38768 col capacity_vph;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38848 col capacity_vph; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 38849 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links)
row 39135 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39192 col capacity_vph;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39193 col capacity_vph; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 39195 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links)
row 39296 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39313 col capacity_vph;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39314 col capacity_vph; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 39342 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links)
row 39343 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153265 col capacity_vph;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153268 col capacity_vph]

Likely cause: Export mapping dropped the column for some records, or the network/matrix file has blank attributes.

[src: findings/3f089cf9e215; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38768 col capacity_vph;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38848 col capacity_vph; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 38849 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links)
row 39135 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39192 col capacity_vph;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39193 col capacity_vph; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 39195 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links)
row 39296 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39313 col capacity_vph;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39314 col capacity_vph; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 39342 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links)
row 39343 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153265 col capacity_vph;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153268 col capacity_vph]

Suggested action: Fill or default capacity_vph in the source file and re-export; downstream checks skip rows with nulls.

[src: findings/3f089cf9e215; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38768 col capacity_vph;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38848 col capacity_vph; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 38849 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links)
row 39135 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39192 col capacity_vph;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39193 col capacity_vph; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 39195 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links)
row 39296 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39313 col capacity_vph;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39314 col capacity_vph; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2025 fields) (links) row 39342 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links)
row 39343 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153265 col capacity_vph;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153268 col capacity_vph]

Evidence

Metric	Value	Threshold
table	links	
column	capacity_vph	

Metric	Value	Threshold
null_rows	14	
rows	136,509	
null_share	0.0%	
example_ids	14 items (full list in findings.json): 38,768, 38,848, 38,849	
issue	null_values	
null_share_high		1.0%
null_share_medium		0.0%

[src: findings/3f089cf9e215; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38768 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38848 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38849 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39135 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39192 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39193 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39195 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39296 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39313 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39314 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39342 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39343 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153265 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153268 col capacity_vph]

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38768 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38848 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38849 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39135 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39192 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39193 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39195 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39296 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39313 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39314 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39342 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39343 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153265 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153268 col capacity_vph

[src: findings/3f089cf9e215; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38768 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38848 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 38849 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39135 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39192 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39193 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39195 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39296 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39313 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39314 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39342 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 39343 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153265 col capacity_vph; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2025 fields) (links) row 153268 col capacity_vph]

Info findings (0)

No Info findings for this run.

Convergence and noise

Check status: skipped; findings: 0; message: required table(s) missing: convergence.
[src: check_results.json]

The convergence check raised no findings.
[src: findings.json]

No convergence table was exported with this run, so iteration history is not available.

No noise band has been computed for this run (derived/noise_band.parquet is missing), so significance tests use the default rule.

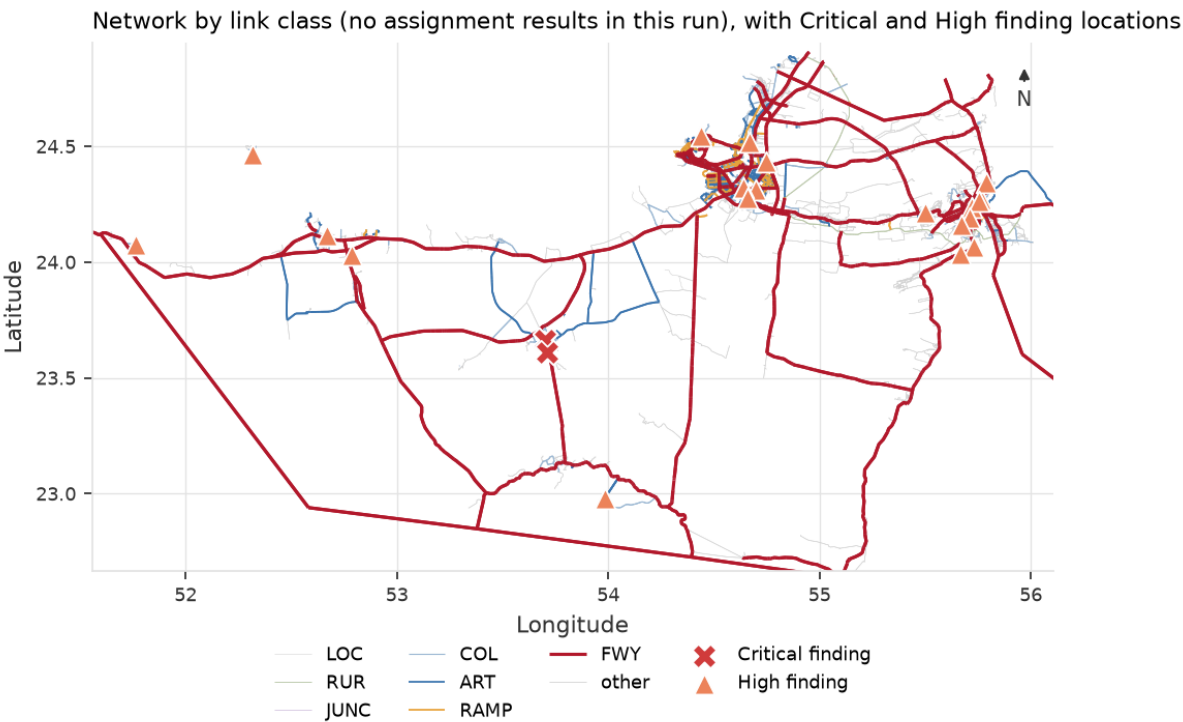
Key performance indicators

KPIs with their published definitions

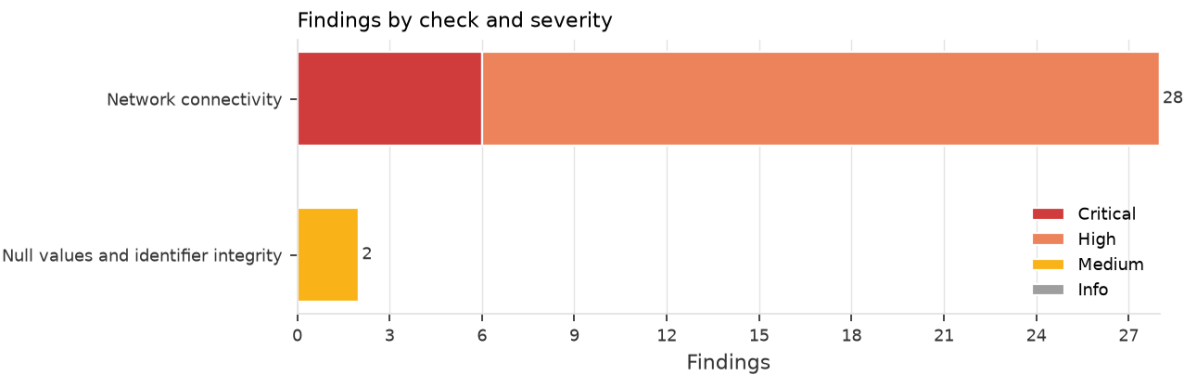
KPI	Value	Period	Definition
Link length, each direction	28279.1 km	all	Sum of link length over all highway links (each direction counted). SQL: SELECT SUM(length_m) / 1000.0 AS value FROM links
Lane-km, all roads	47,736 lane-km	all	Sum of link length x lanes over all highway links. SQL: SELECT SUM(length_m * lanes) / 1000.0 AS value FROM links
Lane-km, freeways	21,802 lane-km	all	Sum of link length x lanes over links of class FWY. SQL: SELECT SUM(length_m * lanes) / 1000.0 AS value FROM links WHERE link_class = 'FWY'

[src: kpis.json; config/kpis.yaml]

Maps and charts



Network map: links by class (the run has no assignment results), Critical and High finding locations marked
[src: links table (geometry_wkt, link_class); findings.json]



Findings by check and severity
[src: findings.json; 30 findings]

Check coverage

13 checks were run: 3 completed, 10 skipped because a required table was missing, 0 ended in error.
[src: check_results.json]

Check coverage

Check	Status	Findings	Message	Duration (s)
Null values and identifier integrity	ok	2		0.11
Network connectivity	ok	28	disconnected_component: 2,343 occurrences grouped into one summary plus the 10 most severe; node_id_mismatch: 73 occurrences grouped into one summary plus the 10 most severe	2.39
Centroid connector placement	ok	0		0.03

Check	Status	Findings	Message	Duration (s)
Transit line integrity	skipped	0	required table(s) missing: transit_lines, transit_segments	0.00
Land-use control totals	skipped	0	required table(s) missing: control_totals, land_use	0.00
Link volume outliers	skipped	0	required table(s) missing: link_flows	0.00
Convergence and noise band	skipped	0	required table(s) missing: convergence	0.00
Demand matrix sanity	skipped	0	required table(s) missing: od	0.00
Trip length distribution	skipped	0	required table(s) missing: od, skims	0.00
Unrealistic speeds and times	skipped	0	required table(s) missing: link_flows	0.00
Parameter drift	skipped	0	required table(s) missing: parameters	0.00
Unexplained demand shift	skipped	0	required table(s) missing: od	0.00
Unused transit services	skipped	0	required table(s) missing: line_loads, od, transit_lines, transit_segments	0.00

[src: check_results.json]

DRAFT

Appendix

Method notes

Each check maps a computed score to a severity band. Bands are checked top-down and the first match wins. The thresholds below are read from config/checks.yaml (defaults: enabled=True, max_findings_per_check=200, group_issue_above=25, group_keep_worst=10).

[src: config/checks.yaml]

Check thresholds and severity rules

Check	Enabled	Parameters	Severity rules
null_and_id_integrity	yes	required_columns = {'links': ['link_id', 'a_node', 'b_node', 'length_m', 'link_class', 'capacity_vph', 'ffs_kph'], 'nodes': ['node_id', 'x', 'y'], 'zones': ['zone_id'], 'link_flows': ['link_id', 'period', 'user_class', 'volume']}	Critical when missing_node_refs > 0; Critical when missing_zone_refs > 0; High when missing_link_refs > 0; High when null_share > 0.01; Medium when null_share > 0
network_connectivity	yes	min_capacity_vph = 1; min_ffs_kph = 5; touch_tolerance_m = 1; near_gap_m = 50	Critical when issue == 'zero_capacity' or issue == 'zero_speed'; High when issue == 'node_id_mismatch'; Medium when issue == 'orphan_node'; High when issue == 'oneway_dead_end'; High when issue == 'disconnected_component'
centroid_connectors	yes	high_class_links = FWY, EXP; connector_classes = CONN; max_connector_length_m = 5,000	High when issue == 'connector_on_high_class'; Medium when issue == 'connector_too_long'
transit_line_integrity	yes	headway_min_range = 2, 120; modes_headway_max = {'METRO': 20, 'LRT': 30, 'BRT': 30, 'BUS': 120, 'RAIL': 120, 'TRAM': 30, 'FERRY': 120}	Critical when issue == 'broken_sequence'; High when issue == 'headway_out_of_range'; Critical when issue == 'node_not_in_network'
land_use_control_totals	yes	tolerance_share = 0.02	Critical when abs_diff_share > 0.10; High when abs_diff_share > 0.05; Medium when abs_diff_share > 0.02
link_volume_outliers	yes	vc_critical = 1.3; vc_high = 1.1; vc_medium = 0.95; zscore_threshold = 3; min_vc_for_outlier = 0.25; min_group_size = 20; low_volume_share_of_group_median = 0.02; user_class = ALL; periods = AM, PM; exclude_link_classes = CONN	Critical when vc_ratio >= 1.30; High when vc_ratio >= 1.10; High when abs_zscore >= 4.0; Medium when vc_ratio >= 0.95 or abs_zscore >= 3.0; Medium when too_low
convergence_and_noise	yes	rel_gap_target = 0.001; rel_gap_high = 0.01; geh_stable = 1; geh_stable_share_target = 0.95; noise_band_iterations = 3; stages = HWY_ASSIGN, PT_ASSIGN	Critical when rel_gap > 0.01; High when rel_gap > 0.001; Medium when stable_share < 0.95
matrix_sanity	yes	intrazonal_share_max = 0.15; row_col_ratio_max = 5; min_zone_total_for_ratio = 50; pa_format_share = 0.25; fractional_cell_max_trips = 0.01; fractional_share_min = 0.01	Critical when issue == 'negative_cells'; High when issue == 'row_col_imbalance'; Medium when issue == 'intrazonal_share'; Info when issue == 'fractional_artefact'
trip_length_distribution	yes	max_mean_diff_share = 0.15; max_mean_diff_share_high = 0.3; distance_bins_km = 0, 2, 5, 10, 20, 40, 80, 200; skim_dist_units = km; reference_mean_km = {}	High when mean_diff_share > 0.30; Medium when mean_diff_share > 0.15
unrealistic_speeds_times	yes	min_cong_speed_kph = 5; max_speed_over_ffs_ratio = 1.05; max_delay_s = 900; periods = AM, PM; user_class = ALL	High when issue == 'speed_above_ffs'; High when issue == 'speed_below_floor'; Medium when issue == 'delay_outlier'
parameter_drift	yes	numeric_tolerance = 1e-09	High when drifted
unexplained_demand_and_shift	yes	sector_shift_share = 0.1; catchment_km = 5; min_sector_pair_trips = 100; explain_land_use_share = 0.01	High when shift_share > 0.25; Medium when shift_share > 0.10
unused_transit_services	yes	min_load_factor = 0.05; min_catchment_demand = 500; catchment_km = 1; pt_modes = PT	High when load_factor < 0.02; Medium when load_factor < 0.05

[src: config/checks.yaml]

Glossary

Glossary

Term	Meaning
V/C	Volume-to-capacity ratio: link volume per hour divided by link capacity per hour. Values above 1.0 mean demand exceeds capacity.
GEH	A statistic comparing two flows, M and C: $\sqrt{2 (M - C)^2 / (M + C)}$. Values below 5 are usually treated as a good match.
Relative gap	Assignment convergence measure: the excess cost of the current route choices over the cheapest possible routes, divided by total cost. Smaller is better.
Noise band	The size of link flow change between the last assignment iterations. Differences inside the band are not treated as significant.
Centroid connector	An artificial link that loads a zone's trips onto the network at a chosen node.
Screenline	An imaginary line across the network on which modelled and observed crossings are compared.

[src: STEAM-AI glossary]

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